Lesson plan · for the teacher

The best plan is at a corner

Optimization · Level 2 · Lesson 2

Time
40 min

Materials
Graph paper, a ruler and a pencil for each group; The garden's three limits written on the board; Coloured chalk for the lines

Learning goals

- Draw each limit of the garden as a line and shade the plans that fit.
- Find the best plan at a corner by sliding the profit line.
- See how a price or a limit change moves the best corner.

Activity: The garden on graph paper

1. Warm-up (5 min): Tell the garden story: tomatoes earn 30 coins a bed, beans 40, and land, water and pupils' hours are limited. Ask pupils to guess the best mix and write a few guesses on the board.
2. Teach (8 min): Draw the axes (tomato beds across, bean beds up). Draw $x + y \le 8$ by joining (8, 0) and (0, 8) and shade towards (0, 0). Explain that each limit is a line and the plans that fit are on one side.
3. Draw (12 min): Groups draw $2x + y \le 14$ (from (7, 0) to (0, 14)) and $x + 3y \le 18$ (from (18, 0) to (0, 6)), then shade where all three hold. Walk round and check the corners: (0, 0), (7, 0), (6, 2), (3, 5), (0, 6).
4. Corners (10 min): Each group works out $30x + 40y$ at every corner and circles the best: (3, 5), 290. Compare with the guesses from the warm-up.
5. What if (5 min): Ask groups to predict what happens when tomatoes sell for 50 coins, then check: the best moves to (6, 2), 380. Ask whether profit should be the garden's only goal.

Let 1 square be 1 bed, from 0 to 18 across and 0 to 14 up, so every line fits. No graph paper? Rule a grid on plain paper or draw one big graph on the board.

Discussion

- Why is the best plan at a corner and not in the middle?
- Why doesn't extra water change the best plan?
- Should profit be the garden's only goal? What else matters?

Exit ticket

- Sketch the shaded shape and mark the best corner. Which two limits meet there?

Try it at school tomorrow
Give each group graph paper and the garden's three limits. Pupils draw the lines, shade the plans that fit and work out the profit at every corner. Ask which corner wins, and whether they would really plant it.

Practice questions · for you

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1 **Mr Okafor's pupils offer 24 hours a week in the garden instead of 18 (the hours limit becomes x + 3y ≤ 24). Does the best plan stay at 3 tomato beds and 5 bean beds?**

◯ No: the profit per bed changes ◯ No: the hours line moves out, so the corners change

◯ Yes: the best corner never moves

2 **The plan 4 tomato beds and 4 bean beds keeps all three limits (x + y ≤ 8, 2x + y ≤ 14, x + 3y ≤ 18).**

◯ True ◯ False

3 **Put the steps of solving a two-variable linear program on graph paper in order.**

☐ Shade where every limit holds

☐ Draw each limit as a straight line

☐ Pick the best corner and check it makes sense

☐ Mark the corners where the lines cross

☐ Work out the goal at every corner

4 **At the best corner (3, 5), which limits are used up exactly? Choose all that apply.**

☐ None: the best plan never uses a limit up ☐ Land: 3 + 5 = 8 beds ☐ Water: 2 × 3 + 5 = 11 cans

☐ Hours: 3 + 3 × 5 = 18 hours

5 **Complete the sentence.**

In a linear program, the best plan is always at a _____ of the _____ region, so the _____ method walks from corner to corner.

Word bank: centre · corner · feasible · greedy · simplex

✎ **Where in your subject could pupils draw a two-variable problem like the garden? What would the limits and the goal be, and who should agree the goal?**

Check your answers

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1 **No: the hours line moves out, so the corners change**

The hours line moves out, so the shape and its corners change: (3, 5) is no longer a corner.

2 **True**

Land: $4 + 4 = 8 \leq 8$. Water: $8 + 4 = 12 \leq 14$.

3 **Top to bottom: 2, 1, 5, 3, 4**

The lines make the shape, the shape gives the corners, and the best plan is one of the corners, so you only need the goal at each corner.

4 **Land: $3 + 5 = 8$ beds; Hours: $3 + 3 \times 5 = 18$ hours**

Land (8 of 8) and hours (18 of 18) are binding: the corner sits where their lines cross.

5 **In a linear program, the best plan is always at a corner of the feasible region, so the simplex method walks from corner to corner.**

The profit line can always slide past a point inside the feasible region, so the best plan sits at a corner.